

Applied Machine Learning
(CINTO2401E)
Group 26

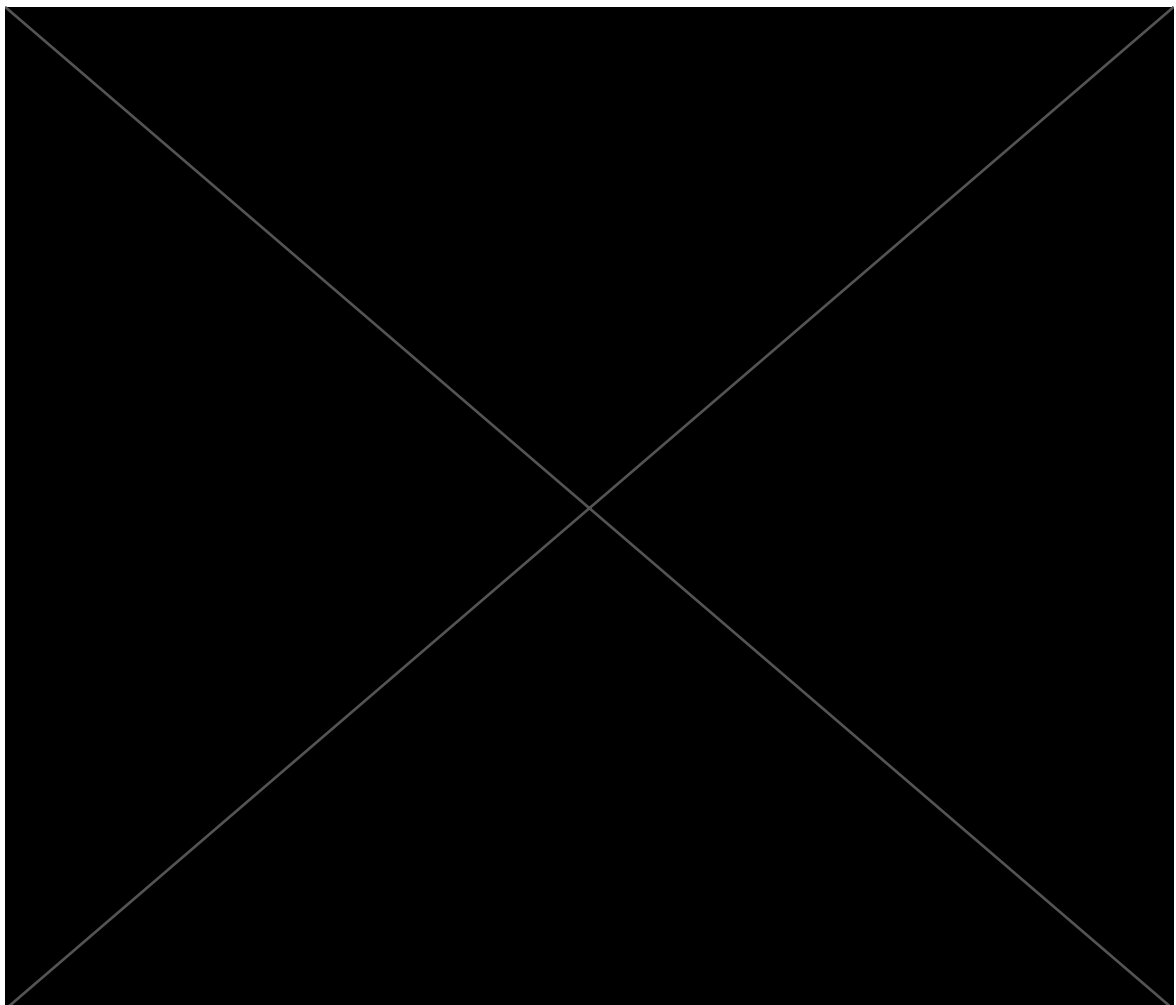


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1 Introduction

1.1 Business Case and Machine Learning Problem

Over the past decade, Airbnb has become one of the most influential and significant actors in the global tourism industry. Last year, the company celebrated the milestone of reaching five million hosts worldwide (Airbnb, 2024; Guttentag, 2019). For many individuals, Airbnb represents an accessible way to generate supplemental income by renting out spare rooms, entire homes, or second homes when it best suits their personal circumstances (Airbnb, 2024).

Despite the platform's intuitive design, determining an optimal nightly rate remains a significant challenge for new hosts (Chami, 2025). In the absence of historical booking data or local market experience, hosts often rely on intuition when pricing their listings, which can result in a disconnect between prices and market demand. Overpricing can lead to high vacancy and missed bookings, whereas underpricing can result in forfeited revenue (Hometime, 2025).

Price is widely recognized as a primary driver of booking performance, yet it is influenced by a complex array of factors including location, amenities, and capacity, making it difficult for hosts to determine a competitive price on their own (Bedspoke, 2016). This project seeks to explore how these factors influence market-accepted pricing levels and provide hosts with an automated pricing recommendation tool. Ideally integrated as the final step in the listing creation process, the tool invokes a machine learning model once a host has entered their property details. Hosts would remain free to adjust the price, but the model aims to reduce reliance on intuition by providing an evidence-based starting point. Such a tool would not only benefit individual hosts but could also improve the broader Airbnb ecosystem by increasing booking performance, minimizing inactive listings, and enhancing overall marketplace efficiency.

To achieve this, the project adopts a predictive modeling approach. The primary objective is to model the relationship between observable listing features and their corresponding market price. Predictive modeling allows forecasting unknown values by learning from patterns, while machine learning provides the methodological framework to capture the complex, potentially non-linear relationship between listing features and listing price (Provost & Fawcett, 2013).

1.2 Use of GenAI

Large Language Models (LLMs), specifically Google's Gemini and OpenAI's ChatGPT, were used for idea generation and conceptualization for this project in accordance with CBS' guidelines (Copenhagen Business School, n.d.).

2 Data

2.1 Data Set and Target Variable

The underlying data set supporting this machine learning project is a point-in-time extract of all Airbnb listings in Copenhagen, Denmark on June 27th, 2025. The raw data includes attributes such as listing name, description, neighborhood, number of beds and bathrooms, how

many guests can be accommodated, and nightly price. The data set was sourced from the publicly available website, [Inside Airbnb](#). Access was obtained via *listings.csv.gz* with an initial size of 22,684 instances (rows) and 79 attributes (columns). This dataset is particularly relevant for this regression task because it contains a wide range of features that are assumed to influence the price a listing commands, which allow the selected models to learn patterns that represent market-driven pricing decisions in a realistic context.

The target variable for this supervised learning task is the nightly price of an Airbnb listing in local currency, DKK (attribute: “price”). This variable represents the price set by a property’s host at the time of data collection and serves as a proxy for market pricing behavior. It is a continuous numeric value.

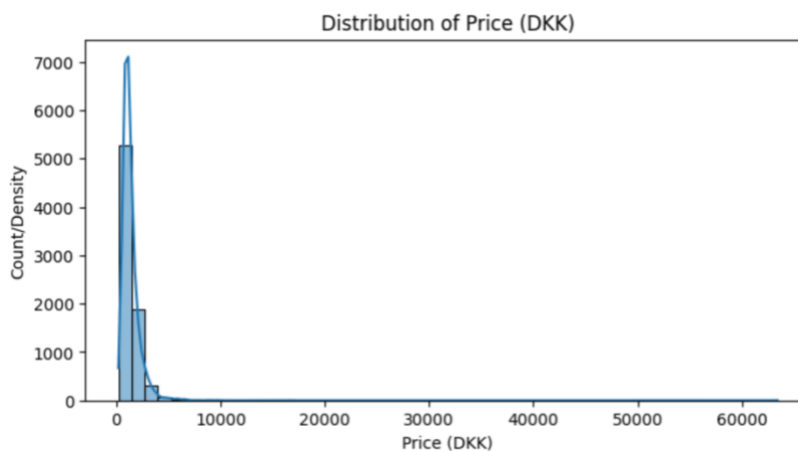


Figure 1: Distribution of price

As shown in the price distribution histogram above, nightly Airbnb prices are concentrated in the lower price brackets (under ~5,000 DKK) with a long “tail” of outliers extending beyond 60,000 DKK. This right skew has several implications for the modeling process, which are discussed in section 2.4.

2.2 Data Cleansing

The raw data set contained 79 attributes and 22,684 instances. The initial data audit revealed null values, incorrect data types, and improperly formatted values. To correct these issues, the cleansing activities outlined in the following sections were taken, which resulted in a final data set consisting of 40 attributes and 7,567 instances. The cleansed data set was used for modeling.

2.2.1 Dropped Attributes

Many attributes were considered irrelevant or redundant and were therefore dropped. For example, attributes such as *listing_url*, *scrape_id*, *source*, and *calendar_last_scraped* represent metadata and are irrelevant for pricing predictions. Some attributes, such as *photo* and non-categorical text columns, were excluded to limit methodological complexity, as activities such as image extraction and text analysis are considered out of scope for this project. For some non-categorical text columns, replacement features were engineered to capture signals (see

section 2.2.5 for more details). All excluded attributes and associated rationale are listed in Appendix 8.1.

2.2.2 *Dropped Rows*

For attributes with a relatively small percentage of missing values, the corresponding rows were dropped rather than imputed, as the data set still contained enough instances to effectively train the models. This included null values for *host_is_superhost*, *host_tenure_days* (an engineered column), *bedrooms*, and *bathrooms*. Missingness was identified in the target variable (*price*) for approximately 20% of observations. Despite the relatively high percentage of null values, imputation was not performed, as it would have caused circular reasoning (i.e., training the model to predict its own best guess).

2.2.3 *Imputations*

To ensure model integrity and prevent data leakage, no inter-row imputations for null values were performed during the initial data cleansing stage. Imputing missing values using statistics derived from the entire dataset, rather than solely the training subset, allows information from the eventual test set to leak into the training data, leading to artificially inflated performance metrics and a false sense of model generalization (Gomede, 2024). Consequently, the model appears more accurate during training than it would be when deployed on truly unseen data. As such, prior to establishing the Scikit-learn pipeline and column transformer, only simple, intra-row imputations were calculated. For instance, the initial data audit revealed that the *beds* attribute contained 2,261 null values. Intra-row imputation was performed under the assumption that there is, on average, one bed per bedroom. The null values of *beds* were therefore imputed with the corresponding number of bedrooms.

2.2.4 *Binary Feature Standardization*

As shown in the initial data audit, several attributes that represent Boolean values were not stored as such but rather encoded as character strings with values such as “t” and “f”. To ensure consistent processing, these values were first mapped to numeric indicators, where “t” was converted to 1 and “f” to 0. The variables were then cast into a proper Boolean data type. This approach ensures compatibility with preprocessing pipelines, while preserving the binary nature of the features.

2.2.5 *Feature Engineering*

As part of the data cleansing and wrangling process, multiple new attributes were created, and various existing attributes were transformed.

The attributes *description*, *host_about*, and *neighborhood_overview* are non-categorical, free text fields. While these attributes are assumed to influence both listing price and booking behavior through perceived quality and professionalism, the natural language processing (NLP) techniques required to make use of them for machine learning purposes fall outside the scope of this project (Belcic, 2025). However, to retain a portion of their influence in the modeling phase, the attributes were transformed into numerical values by measuring the length

of the text, which serves as a proxy for information richness (*description_length*, *host_about_length*, *neighborhood_overview_length*). Furthermore, additional attributes were engineered to indicate missingness, reflecting the assumption that a null description differs from short, but present, descriptions. These engineered features are *description_missing*, *host_about_missing*, and *neighborhood_overview_missing*.

The *host_location* field is not directly relevant to a listing's price but nonetheless contains useful information. The attribute *host_location_missing* was created to replace *host_location* to mitigate the risk of direct bias¹ (a model cannot learn to directly penalize or reward a listing based solely on the value of the host's location), while still allowing the model to learn a separate coefficient for listings where this information is unavailable, capturing the potential price penalty or premium associated with this specific type of missing data.

The *host_tenure_days* column was engineered from the *host_since* column, by calculating the number days between the host's sign-up date and the snapshot date. This new attribute provides the model with a numerical measure of host experience that it can process, unlike the original date format. This was done based on the assumption that host tenure serves as a proxy for trust, reliability, and market experience. A longer host tenure likely correlates with established reputation and higher quality service, which are factors that likely allow a listing to command a premium price.

The attributes *host_response_rate* and *host_acceptance_rate* were supplemented with *host_response_rate_missing* and *host_acceptance_rate_missing* to avoid applying inter-row imputation before the train-test split. The additional attributes consist of binary indicators, taking the value 1 if the original attribute is missing and 0 if not, thereby allowing the model to capture informative missingness. By retaining the original attributes as well, the percentage value (where not null) still reflects the host's performance.

A grid of granular hexagonal bins (transformed into a binary indicator column for each bin) was created from listings' coordinates to complement the *neighborhood_cleansed* attribute, as these predefined neighborhoods cover relatively large geographic areas. In practice, it is presumed that Airbnb demand is often driven by much more detailed location preferences, where small differences in proximity can have a meaningful impact on perceived attractiveness and pricing. By introducing hex-bin-based spatial features, the models aims to capture finer-grained location effects that are not reflected in broad neighborhood categories. It is therefore hypothesized that this spatial representation will improve the model's ability to learn localized price variation and, in turn, overall predictive accuracy.

2.3 Train-Test Split and Preprocessing

Before any preprocessing steps (i.e., one-hot encoding and inter-row imputations) were applied, a train-test split was performed. This divided the data into training and test sets, where

¹ Dropping the column does *not*, however, prevent the model from learning underlying bias (if any) through proxy features (indirect discrimination).

80% was allocated to the training set and the remaining 20% to simulate new and unseen data for final model evaluation. The training set was used for baselining, training, and hyperparameter tuning, while the test set was used exclusively for the model's final performance evaluation. It was crucial to perform the train-test split prior to any preprocessing steps to prevent information from the test set from influencing any decision-making or calculations ("data leakage") (scikit-learn, 2025).

After the train-test split, the training set features were categorized based on data types to enable category-specific pre-processing. This included a pipeline for imputing missing values with the median and performing scaling on the numerical features. Using median over mean for imputation was prudent for attributes like *host_acceptance_rate* and *host_response_rate*, because it ensured less sensitivity to the extreme outliers (Vaj, 2024).

Boolean features were passed through the pipeline during preprocessing, but categorical attributes were transformed using one-hot encoding, which allowed models to learn from the categorical information while avoiding ordinality (GeeksForGeeks, 2025). Simultaneously, to avoid production errors, the encoder was set to ignore unknown categories that were not seen during the training process (GeeksForGeeks, 2025). Transformations were applied to each data category via ColumnTransformer, and data preprocessing was complete.

2.4 Data Limitations

The target variable of price is manually, and perhaps arbitrarily, set by hosts and may not reflect true market-clearing prices. In addition, because the data point-in-time, it contains no information on whether a given price (or any other attributes) resulted in future bookings. However, for a pricing recommendation tool, this does not undermine the approach. By learning from prices of comparable listings, the model still provides a reasonable, market baseline that reflects prevailing pricing norms.

The *amenities* attribute is a free-text field, resulting in a high number of unique values, thereby exploding the dimension count when one-hot encoding and potentially decreasing model performance. As such, it was decided to reduce the amenities list to the top 10 most important amenities for guests, according to Airbnb (Airbnb, 2025). It is acknowledged that this is a significant limitation, since amenities do play an important role in the appeal of a listing, and excluding less common amenities could potentially reduce the model's ability to fully capture variation in listing prices (Airbnb, 2025).

As noted, some attributes and features were dropped based on practical judgment and perceived relevance, which introduces a risk of selection bias (Machine Learning Glossary, n.d.). Efforts were made to mitigate this risk by, for instance, basing the selection of amenities on Airbnb's statistics on most important amenities. In addition, removing attributes due to complexity, such as image-based information, may limit the model's ability to capture certain aspects of listing quality.

Several attributes concerning the hosts, such as name, picture, and other personal identifiers, could potentially influence booking behavior, as studies have shown that such attributes may be associated with discrimination patterns or implicit biases (Morris, 2024). Including these variables could have provided additional explanatory power but raises significant ethical concerns and methodological challenges. As such, these features were intentionally excluded. It is acknowledged that removing these attributes does not ensure impartiality and their indirect effects on booking behavior may still be present in the data (Barocas, Hardt, & Narayanan, 2023)

This project relies exclusively on secondary data, which creates a risk that the available features may not fully capture all the factors that influences the price of an Airbnb listing. In addition, the accuracy and consistency of the secondary data cannot be fully controlled. Some variables could potentially contain measurement errors, outdated information or inconsistencies due to how hosts self-report listing characteristics. Although data cleansing and preprocessing were applied, these potential issues may still affect the model.

3 Modeling

3.1 Learning Algorithms

Supervised learning algorithms are appropriate for inferring relationships between prices and other data attributes, as *price* is a labeled target variable. This project’s objective is to develop a price recommendation tool for new listings, which constitutes a regression problem, since the target variable is a continuous numerical value (James et al., 2023).

By learning the relationship between listing features, such as location, property characteristics, and amenities, and their corresponding prices, the models can estimate an appropriate price for new, unseen listings. This project applies seven different models to account for potential non-linear relationships, varying model assumptions, and differences in bias–variance trade-offs. Incorporating model diversity enables a systematic comparison of predictive performance and supports the identification of the most suitable price recommendation tool.

Model Category	Selected Model(s)	Rationale
Linear	Linear Regression, Lasso, Ridge	- Provides transparent and interpretable results (James et al., 2023) - Lasso and Ridge mitigate multicollinearity and reduce overfitting by applying L1 and L2 regularization, respectively, zeroing out or shrinking less important feature coefficients (James et al., 2023)
Tree-Based	Decision Tree	Can capture non-linear relationships (Hastie, Tibshirani, & Friedman, 2009)

Ensemble	Random Forest, GradientBoosting (GB)	• Random Forest can significantly reduce the high variance (overfitting) of individual trees by aggregating predictions from many decorrelated trees, improving stability and accuracy (Friedmand, 2001) • GB sequentially builds trees, where each new tree corrects the errors of the previous ones (Friedmand, 2001)
Instance-Based	K-Nearest Neighbors (KNN)	• Included as a non-parametric baseline (Kavlakoglu, n.d) • A simple, memory-based method that classifies/regresses a point based on the prices of its “K” nearest neighbors, making it a relevant check for the importance of local similarity in determining price (Kavlakoglu, n.d)

3.2 Establishing Baseline Models

To provide a reference point for model performance before hyperparameter tuning, baseline models were trained and evaluated. Since a single train–test split may be particularly lucky or unlucky, and therefore provide an inaccurate estimate of generalized performance, k-fold cross-validation was employed (Provost & Fawcett, 2013). In this approach, the training data set is split into five equal-sized folds, where one fold is used for validation, while the remaining four folds are used for training. To prevent data leakage, the procedure is applied exclusively to the training data. As noted, a diverse set of baseline models enables comparing different levels of complexity and helps identify models that are suboptimal (in this case, the decision tree algorithm was excluded from GridSearchCV).

3.3 Defining and Tuning Hyperparameters

Following the selection of candidate models, GridSearchCV was utilized to perform systematic hyperparameter optimization. This process began by defining a parameter grid for each estimator, which specified the ranges of hyperparameters to be tested.

GridSearchCV employs k-fold cross-validation for each hyperparameter combination, and in this case, ultimately selecting the configuration that minimizes the Mean Absolute Error (MAE). In the context of a relatively limited dataset, this exhaustive validation is essential for producing robust and reliable accuracy estimates (GeeksForGeeks, 2025).

Once the optimal hyperparameters were identified, a final round of cross-validation was conducted to calculate a comprehensive suite of evaluation metrics, including Root Mean Squared Error (RMSE) and R^2 . This multi-metric assessment ensures a holistic view of each model's performance, balancing average error sensitivity with an understanding of variance and outlier impact (K, 2024).

4 Results

4.1 Training Data GridSearchCV Results

As previously mentioned, hyperparameters for each model were selected using GridSearchCV with 5-folds cross-validation on the training data. The strongest hyperparameter combinations (according to the selected metrics) served as the basis for the final model evaluations on the holdout test set. See appendix 8.2 for GridSearchCV results on training data.

4.2 Evaluation Metrics

Model performance was evaluated using Mean Average Error (MAE), Root Mean Squared Error (RMSE), and R^2 . These metrics were selected to capture complementary aspects of predictive accuracy and together provide a more comprehensive evaluation of model performance (K, 2024).

MAE was selected as the primary evaluation metric because it aligns closely with the intended use case of providing hosts with an intuitive and interpretable pricing recommendation in the same unit of measurement as actual prices. An MAE of approximately 350 DKK represents a typical deviation that is directly understandable in practical terms. A key property of MAE is that all errors contribute linearly to the final score, making it less sensitive to extreme outliers (K, 2024).

RMSE is reported alongside MAE to capture sensitivity to larger prediction errors. By squaring residuals prior to aggregation, RMSE assigns greater weight to large deviations and therefore highlights models that occasionally produce substantial pricing errors, particularly at higher price levels (K, 2024).

R^2 is included as a complementary metric to assess overall explanatory power. It indicates how much of the variation in listing prices is captured by the model and provides a general measure of how well the model fits the data (K, 2024). However, since R^2 is not measured in monetary terms, it is therefore interpreted alongside error-based metrics (K, 2024).

4.3 Test Set Model Performance

Model	Test MAE	Test RMSE	Test R^2
Linear Regression	385.7	563.7	0.55
Ridge Regression	377.0	554.5	0.57
Lasso Regression	378.1	559.9	0.56
Random Forest	345.2	524.6	0.61
Gradient Boosting	348.3	541.8	0.59
K Nearest Neighbors	374.8	601.7	0.49

The performance metrics summarized in the table above reveal clear difference across model categories.

Within the linear category, Linear Regression exhibited the weakest performance. An MAE of 385.7 DKK indicates that the model's nightly price recommendations deviate from actual market rates by ~386 DKK on average. The elevated RMSE relative to the MAE suggests that the model is particularly sensitive to outliers, occasionally producing large pricing errors. Furthermore, an R^2 value of 0.55 demonstrates that the linear baseline explains only 55% of the observed price variance on unseen data. Ridge and Lasso regression yielded marginal improvements over the standard linear model. By incorporating regularization penalties, these models achieved slightly lower average prediction errors and improved stability. However, the gains remain incremental, suggesting that the primary limitation of linear models in this context is failing to capture the non-linear complexities of the Copenhagen rental market.

The ensemble-based models significantly outperformed the linear suite. Gradient Boosting and Random Forest achieved the strongest predictive performance, with MAEs of 348 DKK and 354 DKK, respectively. These models not only reduced the average error but also exhibited greater robustness to outliers, as evidenced by their lower RMSE values. While Random Forest was a top performer, Gradient Boosting provided the most precise absolute accuracy (lowest MAE), indicating its effectiveness in iteratively correcting residual errors.

KNN achieved an MAE of 375 DKK, outperforming linear models, but it recorded the weakest overall performance in terms of RMSE and R^2 (0.49). This significant gap between MAE and RMSE indicates that while KNN is accurate for "typical" listings, it is highly susceptible to massive errors when faced with sparse data points or outliers.

Across all models, the differences in MAE were accompanied by differences in sensitivity to larger errors, as reflected in the RMSE. The models that had lower average prediction (MAE) also tended to exhibit higher R^2 , which can indicate a closer alignment between predictive accuracy and explanatory power.

4.4 Residual Analysis

Homoscedasticity occurs when there is a uniform variance of residuals across the entire range of predicted values, meaning there is no increase or decrease in trends in terms of the difference between actual and predicted values across price levels. If there are changes in the variance, this is referred to as heteroscedasticity, which can indicate underlying flaws in a model (Santos, 2024). To ensure legitimacy and trustworthiness of the model results, residual plots were created to analyze the homoscedasticity (or lack thereof) across the models (see Appendix 8.3).

While the residual plots initially suggest heteroscedasticity, the interpretation is nuanced by the underlying data distribution. The "fan shape" is partially a result of the high concentration of listings in the lower price range (see section 2.1). Naturally, the absolute scale of errors increases as the target values grow. Importantly, however, the residuals remain relatively symmetric around the zero-line across the entire predicted range. That is, the models aren't biased to predicting higher-than-actual prices vs. lower-than-actual, or vice-versa. This indicates that the models are not systematically biased, since they are equally likely to over-

predict as they are to under-predict. The increased variance at the higher price range suggests that the listing prices become more volatile and less dependent on standard features as the price becomes larger.

The extreme residuals at the high end could indicate either “unobservable” features (i.e., something absent in the data that justifies a higher price) or “unreasonable” price setting by hosts. Spot checks performed on the data indicate the latter. For instance, the most expensive listings in the cleansed dataset (40,000–65,000 DKK) possess features, such as guest capacity and bed count, that are nearly identical to mid-range listings. In these instances, a high residual is actually a sign of model robustness instead of a flaw, since the model is identifying a price based on market fundamentals, rather than being skewed by anomalous data points. For a price recommendation tool, this “error” is the primary value proposition: identifying where a listing is significantly mispriced relative to its peers.

5 Discussion

5.1 Interpretation of Results

The findings of this study indicate that machine learning models based on observable listing characteristics are capable of capturing meaningful pricing patterns in the Airbnb market. The consistent performance gap between ensemble-based models and simpler approaches highlights the importance of accounting for complex, non-linear relationships and interactions between features when modeling short-term rental prices.

The superior performance of tree-based ensemble methods suggests that pricing dynamics are not driven by individual attributes in isolation, but rather by combinations of factors such as location, capacity, and amenities. By aggregating multiple decision trees or iteratively correcting prediction errors, ensemble models are better equipped to adapt to such complexity, whereas linear models remain constrained by their additive structure and underlying assumptions.

Although regularization improves the stability of linear models, the relatively modest gains observed indicate that controlling model complexity alone is insufficient when the true data-generating process is highly non-linear. This limitation is further emphasized by the comparatively weaker performance of instance-based methods, where reliance on local similarity appears less effective in a heterogeneous and high-dimensional feature space.

From a practical perspective, the results suggest that ensemble-based models are more suitable for use as price recommendation tools, as they balance predictive accuracy with robustness to extreme pricing errors. However, the remaining prediction error and unexplained variance also underscore the role of factors not captured in the dataset, such as subjective listing quality, temporal demand fluctuations, and host-specific behavior.

Overall, these findings highlight a trade-off between model complexity, interpretability, and predictive performance.

5.2 Interpretation of Evaluation Metrics

While MAE provides an intuitive measure of typical pricing error, its linear treatment of all deviations means that it does not distinguish between moderate and extreme mispricing. As a result, MAE alone may understate the practical impact of occasional large errors, particularly for high-value listings.

RMSE addresses this limitation by penalizing larger errors, making it effective in identifying models that exhibit unstable behavior at the upper end of the price distribution. However, in a right-skewed market such as short-term rentals, RMSE may be disproportionately influenced by a small number of extreme observations, reducing its representativeness for the average host experience. The consistent gap observed between MAE and RMSE across models therefore suggests the presence of occasional large residuals, primarily driven by high-priced listings, and underscores the importance of interpreting RMSE in conjunction with MAE.

R^2 offers a high-level indication of how well price variation is captured, but it is less informative for practical decision-making. Because it reflects variance rather than absolute error, R^2 does not directly convey how accurate individual price recommendations are in monetary terms.

Taken together, the combined use of MAE, RMSE, and R^2 highlights important trade-offs between average accuracy, robustness to extreme errors, and explanatory power. Within this framework, the Random Forest model emerges as the most balanced candidate for a practical pricing recommendation system.

5.3 Limitations and Ethical Considerations

Several limitations should be acknowledged. First, the dataset is a single snapshot in time and contains no information on whether a given price resulted in bookings or revenue. For practical deployment, the model would need to incorporate historical seasonal data or be continuously retrained on the most recent observations.

Feature selection and transformation choices also introduce limitations. Some attributes were excluded based on scope and complexity, which may introduce selection bias. In particular, the amenities attribute was reduced from a high-dimensional free-text field to a subset of the most important amenities, which may underrepresent less common but price-relevant features. Similarly, transformations such as text-length proxies for descriptive fields provide only coarse representations of richer qualitative information, and the exclusion of complex features such as image-based attributes may limit the model's ability to fully capture perceived listing quality. Ethical considerations further constrain the choice of features. Certain host-related attributes, such as names or profile images, may influence booking behavior through implicit bias. Although including these variables could improve predictive performance, they raise ethical concerns and were therefore intentionally excluded. However, indirect proxy effects may still remain.

Finally, reliance on scraped secondary data introduces uncertainty regarding data quality, consistency, and completeness. Despite extensive data cleansing, self-reported or outdated information may still affect the model's results and should be considered when interpreting its practical applicability.

6 Conclusion

This project examined how observable Airbnb listing characteristics influence market-accepted pricing levels and developed a machine learning based pricing recommendation tool to support hosts in setting competitive nightly rates. Rather than prescribing prices, the aim was to provide an evidence-based starting point that reduces reliance on intuition while preserving host autonomy.

Using a point-in-time dataset of Airbnb listings in Copenhagen from June 27, 2025, the study framed the task as a supervised regression problem with nightly price as the target variable. The dataset proved well suited for this purpose, as it contains a broad set of features related to location, capacity, amenities, and host characteristics that plausibly shape pricing decisions.

A diverse set of baseline models was evaluated to assess different levels of complexity and identify the most suitable model class for the task. The results show that non-linear ensemble methods, particularly Random Forest, outperform linear and instance-based models, indicating that Airbnb pricing is driven by complex, non-linear relationships between listing attributes. While linear models offer interpretability, their weaker performance suggests limited ability to capture these dynamics.

Overall, the findings indicate that machine learning-based price recommendations can provide meaningful decision support for Airbnb hosts, and the approach developed in this project is well suited as a conceptual framework for such a tool. The best-performing model delivers a reasonable benchmark price that is robust to anomalous or idiosyncratic host pricing, making it valuable as a starting point for host decision-making.

However, the model in its current form should not be considered ready for direct implementation. Key limitations include the use of point-in-time snapshot data, reliance on host-listed rather than realized booking prices, and a restricted feature set that omits certain richer textual or visual information. Addressing these limitations would require incorporating time-series data, observed booking outcomes, seasonal effects, and more advanced feature representations, as well as continuous model monitoring and retraining. With these extensions, the proposed approach could evolve from a proof of concept into a production-ready pricing recommendation system.

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8 Appendix

8.1 Dropped Attributes

Feature	Keep	Rationale and notes
listing_url	FALSE	Irrelevant.
scrape_id	FALSE	Irrelevant.
last_scraped	FALSE	Irrelevant – but used to create host_tenure_days.
source	FALSE	Irrelevant.
name	FALSE	Intuitively, the name could impact a listing's "success," but due to the complexity and the scope of this project, we are dropping it.
description	FALSE	Taking the character count to see if the length of the description has an impact on availability. This information is in the new "description_length" and "description_missing" attributes.
neighborhood_overview	FALSE	Taking the character count to see if the length of the host's neighborhood description has an impact on availability. This information is in the new "neighborhood_overview_length" and "neighborhood_overview_missing" attributes.
picture_url	FALSE	Intuitively, the picture would likely have an impact on a listing's "success," but due to the complexity and the scope of this project, we are dropping it. There is also no missingness in this column, so analyzing whether or not it is present is irrelevant.
host_id	FALSE	Irrelevant.
host_url	FALSE	Irrelevant.
host_name	FALSE	Intuitively, the host's name could impact a listing's "success," (discrimination, etc.) but due to the complexity and the scope of this project, we are dropping it.
host_since	FALSE	We have transformed this column into "host_tenure_days," which calculates the number of days they have been a host.
host_location	FALSE	We have transformed this column into "host_location_missing," which indicates whether or not it is missing. Intuitively, the actual host location could impact a listing's "success," (discrimination, etc.) but due to the complexity and the scope of this project, we are dropping it.
host_about	FALSE	Taking the character count to see if the length of the host's "about" field has an impact on availability. This information is in the new "host_about_length" and "host_about_missing" attributes.
host_response_time	TRUE	Added a category called "unknown" if a value is missing.
host_response_rate	TRUE	
host_acceptance_rate	TRUE	
host_is_superhost	TRUE	Transformed to boolean, and where value is missing, set to 0 (not superhost).
host_thumbnail_url	FALSE	Intuitively, the host's thumbnail could impact a listing's "success," (discrimination, etc.) but due to the complexity and the scope of this project, we are dropping it.
host_picture_url	FALSE	Intuitively, the host's picture could impact a listing's "success," (discrimination, etc.) but due to the complexity and the scope of this project, we are dropping it.
host_neighbourhood	FALSE	Irrelevant, really high missingness.
host_listings_count	FALSE	Dropping this column in favor of "calculated_host_listings_count," because that is the direct calculation of how many listings a host has at the time of scrape. The metadata also notes that this calculation is "unknown," and thereby less trustworthy.
host_total_listings_count	FALSE	Dropping this column in favor of "calculated_host_listings_count," because that is the direct calculation of how many listings a host has at the time of scrape. The metadata also notes that this calculation is "unknown," and thereby less trustworthy.

host_verifications	FALSE	Dropping in favor of "host_identity_verified," which is a boolean that indicates whether or not the host is verified.
host_has_profile_pic	TRUE	
host_identity_verified	TRUE	
neighbourhood	FALSE	Dropping in favor of "neighborhood_cleansed."
neighbourhood_cleansed	TRUE	Going to compare to the bins created from "latitude" and "longitude" to see which has a larger impact on availability.
neighbourhood_group_cleansed	FALSE	Dropping, empty attribute.
latitude	FALSE	Used to create hex bins
longitude	FALSE	Used to create hex bins
property_type	TRUE	Kept and created dummy variables for the various categories.
room_type	TRUE	Kept and created dummy variables for the various categories.
accommodates	TRUE	
bathrooms	TRUE	Where this value is missing, take the numeric value from the "bathrooms_text" column and impute.
bathrooms_text	FALSE	Dropping for reason above.
bedrooms	TRUE	
beds	TRUE	
amenities	TRUE	Due to the large amount of values, we have chosen some intuitively important amenities and created dummy variables for them. We recognize that this introduces bias, but for practical reasons, this is the approach we have taken. We also recognize that the specific user would have specific preferences.
price	TRUE	Target variable.
minimum_nights	TRUE	
maximum_nights	TRUE	
minimum_minimum_nights	FALSE	Dropping in favor of "mini_nights" and "maximum_nights"
maximum_minimum_nights	FALSE	Dropping in favor of "mini_nights" and "maximum_nights"
minimum_maximum_nights	FALSE	Dropping in favor of "mini_nights" and "maximum_nights"
maximum_maximum_nights	FALSE	Dropping in favor of "mini_nights" and "maximum_nights"
minimum_nights_avg_ntm	FALSE	Dropping in favor of "mini_nights" and "maximum_nights"
maximum_nights_avg_ntm	FALSE	Dropping in favor of "mini_nights" and "maximum_nights"
calendar_updated	FALSE	Dropping because this attribute is completely empty.
has_availability	FALSE	Dropping because there is no information about what this attribute means.
availability_30	FALSE	Dropping because this attribute will not be available at the time of prediction.
availability_60	FALSE	Dropping because this attribute will not be available at the time of prediction.
availability_90	FALSE	Dropping because this attribute will not be available at the time of prediction.
availability_365	FALSE	Dropping because this attribute will not be available at the time of prediction.
calendar_last_scraped	FALSE	Irrelevant.
number_of_reviews	TRUE	Even though this attribute will be unavailable at the time of prediction for NEW listings, we can set it to an appropriate default of 0 without skewing the results. However, listings that are being "re-activated" WILL have reviews, which intuitively will impact their "success."
number_of_reviews_ltm	FALSE	Dropping in favor "number_of_reviews" for simplicity.
number_of_reviews_l30d	FALSE	Dropping in favor "number_of_reviews" for simplicity.

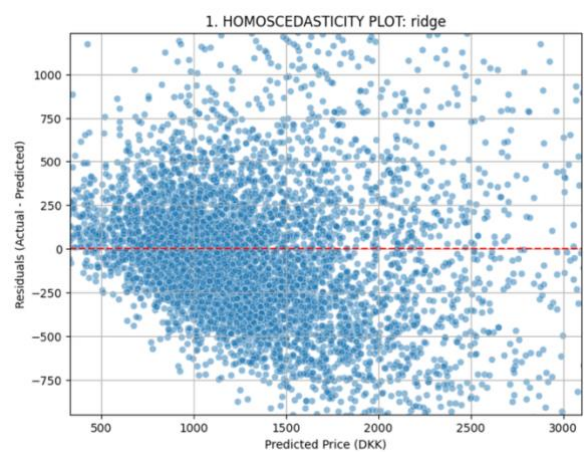
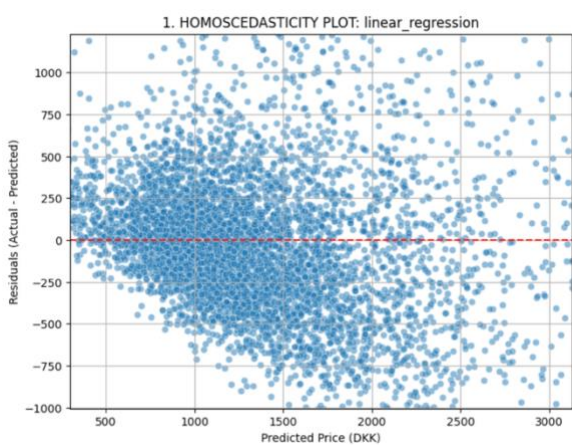
availability_eoy	FALSE	Dropping because there is no information about what this attribute means.
number_of_reviews_ly	FALSE	Dropping in favor "number_of_reviews" for simplicity.
estimated_occupancy_1365d	FALSE	We have chosen to filter this column for listings that have a nonzero value. We have done this to ensure we are only including listings that are active, and thereby more informative for our model. It is not used for training.
estimated_revenue_1365d	FALSE	Irrelevant and would introduce data leakage.
first_review	FALSE	Irrelevant.
last_review	FALSE	We have chosen to filter this column for listings that have a review within 6 months of the scrape date. We have done this to ensure we are only including listings that are active, and thereby more informative for our model. Not used for training.
review_scores_rating	FALSE	Dropping because this attribute will not be available at the time of prediction.
review_scores_accuracy	FALSE	Dropping because this attribute will not be available at the time of prediction.
review_scores_cleanliness	FALSE	Dropping because this attribute will not be available at the time of prediction.
review_scores_checkin	FALSE	Dropping because this attribute will not be available at the time of prediction.
review_scores_communication	FALSE	Dropping because this attribute will not be available at the time of prediction.
review_scores_location	FALSE	Dropping because this attribute will not be available at the time of prediction.
review_scores_value	FALSE	Dropping because this attribute will not be available at the time of prediction.
license	FALSE	Dropping because it's empty.
instant_bookable	TRUE	Transformed to boolean. No missingness.
calculated_host_listings_count	TRUE	
calculated_host_listings_count_entire_homes	FALSE	Dropping in favor of "calculated_host_listings_count" for simplicity.
calculated_host_listings_count_private_rooms	FALSE	Dropping in favor of "calculated_host_listings_count" for simplicity.
calculated_host_listings_count_shared_rooms	FALSE	Dropping in favor of "calculated_host_listings_count" for simplicity.
reviews_per_month	FALSE	Dropping in favor of "number_of_reviews."
description_missing	TRUE	Added feature. See "description" column for explanation.
description_length	TRUE	Added feature. See "description" column for explanation.
neighborhood_overview_length	TRUE	Added feature.
neighborhood_overview_missing	TRUE	Added feature.
host_tenure_days	TRUE	Added feature. See "host_tenure" column for explanation.
host_location_missing	TRUE	Added feature.
host_since_missing	TRUE	Added feature. See "host_since" column for explanation.
host_about_missing	TRUE	Added feature. See "host_about" column for explanation.
host_about_length	TRUE	Added feature. See "host_about" column for explanation.
host_response_rate_missing	TRUE	Added feature. See "host_response_rate" column for explanation.
host_acceptance_rate_missing	TRUE	Added feature. See "host_acceptance_rate" column for explanation.
hex_id	TRUE	Added feature.

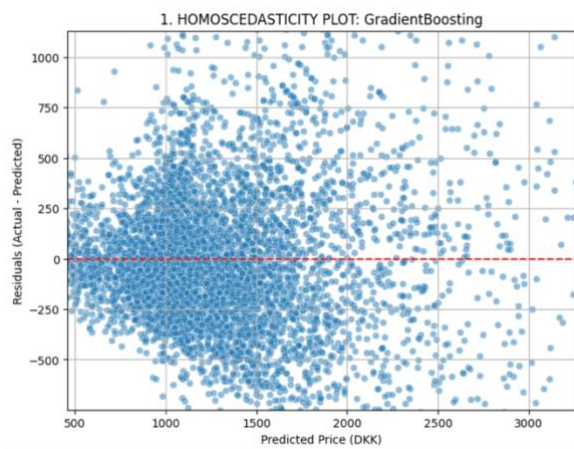
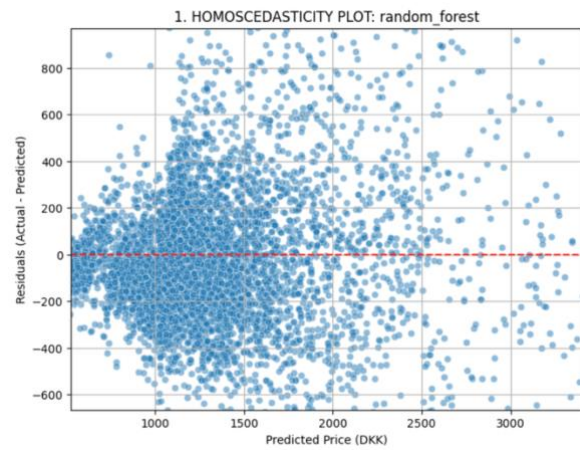
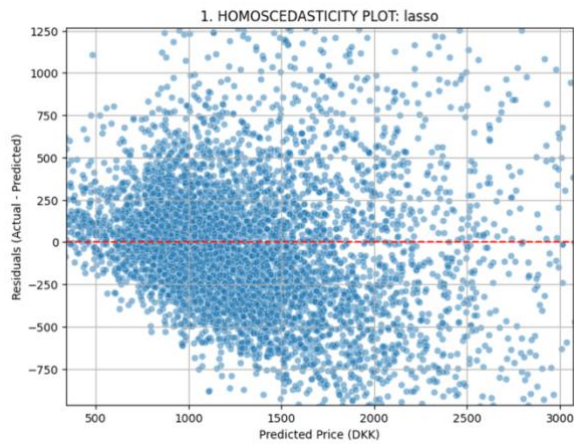
8.2 GridSearchCV Results on Training Data

MODEL	HYPERPARAMETERS	METRICS
LINEAR REGRESSION	N/A	Best CV MAE: 392.5 Best CV RMSE: 1103.5 Best CV R ² : 0.34
RIDGE REGRESSION	{'model__alpha': 10.0}	Best CV MAE: 381.2 Best CV RMSE: 1096.8 Best CV R ² : 0.35
LASSO REGRESSION	{'model__alpha': 1.0}	Best CV MAE: 382.2 Best CV RMSE: 1098.2 Best CV R ² : 0.35
RANDOM FOREST	{'model__max_depth': 20, 'model__min_samples_leaf': 5, 'model__n_estimators': 200}	Best CV MAE: 366.2 Best CV RMSE: 1100.9 Best CV R ² : 0.34
GRADIENT BOOSTING	{'model__learning_rate': 0.1, 'model__n_estimators': 300}	Best CV MAE: 359.6 Best CV RMSE: 1115.9 Best CV R ² : 0.29
K NEAREST NEIGHBORS	{'model__metric': 'manhattan', 'model__n_neighbors': 15, 'model__weights': 'distance'}	Best CV MAE: 383.9 Best CV RMSE: 1123.1 Best CV R ² : 0.29

8.3 Homoscedasticity Plots

TRAINING DATA





TEST DATA

